

Article

Transmission and Generation Expansion Planning Considering Virtual Power Lines/Plants, Distributed Energy Injection and Demand Response Flexibility from TSO-DSO Interface

Flávio Arthur Leal Ferreira ^{1,*} , Clodomiro Unsihuay-Vila ¹  and Rafael A. Núñez-Rodríguez ² 

¹ UFPR—Department of Electrical Engineering, Universidade Federal do Paraná, Curitiba 81531-980, Brazil; clodomiro.vila@ufpr.br

² School Electronic Engineering, Unidades Tecnológicas de Santander, Bucaramanga 680005, Colombia; rnunez281@unab.edu.co

* Correspondence: flavio.ferreira@ufpr.br

Abstract: This article presents a computational model for transmission and generation expansion planning considering the impact of virtual power lines, which consists of the investment in energy storage in the transmission system as well as being able to determine the reduction and postponement of investments in transmission lines. The flexibility from the TSO-DSO interconnection is also modeled, analyzing its impact on system expansion investments. Flexibility is provided to the AC power flow transmission network model by distribution systems connected at the transmission system nodes. The transmission system flexibility requirements are provided by expansion planning performed by the connected DSOs. The objective of the model is to minimize the overall cost of system operation and investments in transmission, generation and flexibility requirements. A data-driven distributionally robust optimization-DDDRO approach is proposed to consider uncertainties of demand and variable renewable energy generation. The column and constraint generation algorithm and duality-free decomposition method are adopted. Case studies using a Garver 6-node system and the IEEE RTS-GMLC were carried out to validate the model and evaluate the values and impacts of local flexibility on transmission system expansion. The results obtained demonstrate a reduction in total costs, an improvement in the efficient use of the transmission system and an improvement in the locational marginal price indicator of the transmission system.

Keywords: expansion planning; energy storage; virtual power line; data-driven distributionally robust optimization



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1. Introduction

1.1. Background

The problem of planning the expansion of electric power transmission and generation systems continues to be the subject of research in the field of electrical engineering. In recent decades, the operational and commercial model of the electricity sector has been restructured on the world stage, bringing new processes, variables and technologies. The current context has evolved towards a situation with the participation of private investors in the areas of commercialization, distribution, transmission and generation, with a reality of competition in the provision of these services. Consumers also began to make their own investments in energy generation, both for consumption and for sale. The expansion planning process must minimize costs, and the variable profit and opportunity for the participating

agents becomes present, as does the need to enable competition in the provision of services, to obtain adequate prices for consumers. Ensuring the reliability and security of the system must be sought with a robust infrastructure to tolerate uncertainties.

Environmental issues related to global warming have restricted the activation of new generation structures that allow deterministic production. The various technologies that have been developed in pursuit of this objective have led, among others, to the growth in the use of renewable energy as a source for electricity generation units. Most of these technologies are characterized by high variability and limited predictability and control. They are often not dispatched by system operators and typically produce energy at a very low marginal cost.

This paper introduces a novel optimization model for expanding generation and transmission infrastructure, using an original modeling and application of battery energy storage to operate in coordination with the load profile of transmission lines, considering storage capacity as an alternative to investing in the expansion of traditional transmission lines. The proposed model also considers the use of variable renewable energy to meet demands, and aggregated energy, power capacity and flexibility, provided to the transmission operation of the power system.

1.2. Literature Review

The operation, planning, business model and regulatory aspects of an electrical system that has a significant presence of generation units with stochastic behavior, such as some renewable sources, are greatly impacted when compared to those used in a system that uses generation that has fully deterministic behavior and is dispatchable [1].

With the context described, the planning of the expansion of the transmission system is increasingly linked to the way in which the expansion of the generation system takes place, leading to the need to consider methods of jointly dealing with the planning of the expansion of these two systems [2].

Planning the capacity expansion of transmission and generation systems is an optimization problem, which needs to have a long-term view, usually analyzed in stages or stages over the proposed time horizon. Bibliographic reviews on the subject can be obtained at [3–5].

In [6], the authors develop a proposed model for the coordinated planning of transmission and distribution expansion. It considers wind generation uncertainties and proposes a 1-stage model, assuming a predefined generation capacity value. It proposes a model that considers a coordinated expansion plan between transmission and distribution, which enables a mutual interest of an optimized investment.

In the work [7] the authors address a model proposal for transmission expansion planning considering the distributed generation of investors to be signaled for the planning process. The proposal generates a distributed energy resources (DER) activation plan (location and capacity) with financial information for investor remuneration. The impact on the locational marginal pricing (LMP) referring to periods of high injection of DER power is considered.

In [8], the authors address a model proposal for generation expansion planning (GEP), considering distributed generation and regulated systems, with centralized planning and operation. A model is proposed for situations where transmission and distribution are centralized and owned by the same company.

In [9], the authors address a model proposal for planning the expansion of the transmission system considering the distributed generation of investors to be signaled for the planning process. The demand and capacity uncertainties of the distributed generators to be signaled are considered. Uncertainties related to natural accidents are considered.

A joint plan for transmission expansion and DER activation (location and capacity) is generated, with financial information for investor remuneration.

In [10], a proposed model for planning the expansion of the transmission system is discussed. It is considered to be a restriction regarding spinning reserves for security in the scenario with renewable energy without inertia. The DDDRO technique is used.

DDDRO generates less conservative solutions than other robust approaches whenever there is an increase in the amount of information from the historical series used to model uncertainty sets. Comparing the DDDRO to stochastic approaches, reference [10] shows the advantages of DDDRO; the authors show that for cases having little historical data available, the DDDRO would provide more conservative solutions compared to stochastic approaches, but, if the available historical data represent the real probability distribution, then the solutions from DDDRO and SP would be the same; however, DDDRO requires less computational time.

The use of the transmission system by the generation, distribution and consumption agents must be charged to make the investments and operating expenses of this system viable. With the aim of establishing a way to allocate these costs among users, methods based on several principles have already been proposed, and what has been used in current models is the so-called nodal method or locational marginal pricing (LMP) [11].

Several exact methods have been used, such as linear programming, nonlinear programming and mixed integer programming, Benders decomposition [12,13] and hierarchical decomposition, solving the integer problem with the [14,15] enumeration algorithm. More recent works that use a robust optimization approach have used a three-level modeling proposal as a reference according to [16,17]:

$$\min_y (c^t y + \max_{u \in U} \min_{x \in \Omega(y, u)} b^t x) \quad (1)$$

where, at the first level of (1), y is the vector with the binary variables referring to investments in transmission lines, generation units and energy storage, and c is the vector of investment costs. At the second level, u is the vector of variable and uncertain generation, and U is the set of uncertainties. At the third level, b is the vector of operating costs and x the vector of operating variables, such as line fluxes, generator dispatch, bus voltage phase angles and load shedding. $\Omega(y, u)$ is the viability region created by operational constraints.

This modeling of the problem can be seen in three layers. At the first level, a better expansion plan is chosen, and the binary investment variables y are identified. At the second level, the worst realization (highest generation and generation curtailment costs) of uncertainties u is considered, considering the feasible scenarios (demand and generation). At the third level, taking the values of y and u as input, the vector x is determined with the operating values that optimize operating costs. Details of this modeling can be found at [18].

Works can be found addressing the modeling of power flexibility from DER, and its use for the benefit of the grid electrical system has been reported in the literature [19–22]. No studies were found considering expansion models of transmission and generation systems to address the impact of the use of ESS at the transmission level, implementing the virtual power line concept, with the aim of postponing or avoiding investments in transmission infrastructure.

Table 1 presents a comparison of the works with the models and solution proposals presented in this review, as well as the proposal being made in this article, considering the characteristics of the planning models.

Table 1. Comparison of the existing approaches with the proposed method.

Ref ¹	GEP ²	TEP ³	UC ⁴	VPP ⁵	VPL ⁶	Flx ⁷	Gen Flx ⁸	DR E Flx ⁹	DR C Flx ¹⁰	T Scale ¹¹	VRE ¹²	Cong ¹³	AC ¹⁴	DC ¹⁵	Sen ¹⁶	Static ¹⁷	Dynamic ¹⁸
[23]	✓		✓				✓				✓	✓				✓	
[24]		✓									✓			✓			✓
[25]		✓											✓			✓	
[26]													✓				
[27]													✓				
[28]		✓											✓			✓	
[29]		✓											✓			✓	
[30]		✓											✓			✓	
[31]	✓		✓				✓				✓	✓		✓		✓	
[32]	✓	✓	✓				✓										
[33]	✓		✓				✓				✓					✓	
[34]	✓	✓	✓				✓				✓			✓		✓	
[35]	✓	✓					✓				✓	✓		✓		✓	
[36]	✓	✓									✓			✓		✓	
[37]	✓	✓					✓				✓			✓		✓	
[7]		✓		✓		✓					✓			✓		✓	
[10]		✓		✓							✓		✓			✓	
[9]		✓		✓		✓					✓			✓		✓	
[8]	✓	✓		✓		✓					✓			✓		✓	
[38]	✓	✓									✓			✓			✓
[39]	✓	✓								✓	✓			✓		✓	
[40]										✓	✓						
[6]		✓				✓										✓	
[41]	✓		✓							✓	✓			✓			✓
[29]		✓											✓			✓	
[42]	✓	✓					✓	✓			✓		✓			✓	
[43]		✓									✓		✓				✓
[44]		✓									✓			✓		✓	
[45]		✓												✓		✓	
[46]		✓					✓							✓		✓	
[47]		✓												✓		✓	
[48]		✓									✓			✓			✓

Table 1. Cont.

Ref ¹	GEP ²	TEP ³	UC ⁴	VPP ⁵	VPL ⁶	Flx ⁷	Gen Flx ⁸	DR E Flx ⁹	DR C Flx ¹⁰	T Scale ¹¹	VRE ¹²	Cong ¹³	AC ¹⁴	DC ¹⁵	Sen ¹⁶	Static ¹⁷	Dynamic ¹⁸
[49]		✓					✓				✓			✓			✓
[50]		✓									✓			✓			✓
[51]		✓					✓			✓	✓			✓			
[52]	✓	✓					✓							✓			✓
[53]		✓	✓				✓				✓			✓			
[54]															✓		
Proposed model	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓		✓

1—Reference, 2—generation expansion planning, 3—transmission expansion planning, 4—short-term unit commitment, 5—virtual power plants, 6—virtual power lines, 7—flexibility TSO-DSO, 8—generation flexibility, 9—DR energy flexibility, 10—DR capacity flexibility, 11—multi-timescale, 12—VRE, 13—transmission line congestion, 14—AC power flow model, 15—DC power flow model, 16—sensitivity analysis, 17—static, 18—dynamic.

1.3. Contributions

This work proposes a model for planning the expansion of transmission and generation systems, taking into account energy storage systems deployed at transmission level implementing virtual power lines concept, variable renewable energy (VRE) injection and flexibility provided at the transmission system operator (TSO) and distribution system operator (DSO) interconnection, through demand response.

The contributions of the paper are as follows:

1. Battery energy storage modeling for implementation of virtual power lines, in generation and transmission expansion planning;
2. Modeling of virtual power plants providing aggregated energy and power capacity to transmission nodes;
3. Modeling of the distributed energy resources services at the TSO-DSO interconnection as demand response flexibility, providing energy and capacity reserve to the transmission system;
4. Implementation of a net demand model associated with load duration curve stages to deal with the use of variable renewable energy.

The rest of the paper is organized as follows: Section 2 presents the problem formulation and models proposed for a deterministic approach, Section 3 presents the uncertainty modeling approach, Section 4 summarizes the solution procedures, case studies are presented in Section 5, and Section 6 concludes this paper. This document ends with the bibliographical references used.

2. Problem Formulation—Deterministic Model

2.1. Net Demand Model

In this paper, the transmission and generation expansion planning (TGEP) proposal considers that there are two types of demand met by the transmission system. In the first type of demand, whose service planning responsibility is a function of a centralized generation expansion plan that is being carried out, with dispatchable and non-dispatchable generation, both existing demands and candidates for installation upon investment are considered in the plan costs.

A second type of demand, which shares the use of the same transmission system, is served by virtual power plants (VPP) contracted by consumers, who provide both the dispatchable generation and non-dispatchable generation needed. The amount of this second type of demand must be compatible with the amount of generation provided by those VPPs.

Demands are modeled using the load duration curve, which shows the relationship between the cumulative load and the percentage of time for which that load occurs. The planning model uses the demand for a given time interval divided into stages, related to load block curves, which breaks down the expected load levels into discrete time blocks. Each time block of a forecasted load demand block is called a stage. The concept of demand is extended to net demand, that is, the demand for electricity minus the contribution from VRE injection. Considering this net demand, the traditional demand duration curve, discretized into four average demand levels (S1, S2, S3, and S4) is presented in Figure 1. Each of the net demand load levels has an associated duration and depth value (in relation to the average net demand load). The sum of the product between the durations and depths must be equal to 1, in order to maintain the average net demand.

For long-term planning purposes, the used net demand stages model is a more coherent assessment of how much each type of generation source is more or less adequate for

the behavior of a load profile. Sources with a generation profile closer to the load profile tend to be more competitive in relation to the others.

In addition, the load stages with lower demands (potential excess generation) are more suitable for signaling the VPL battery charge/discharge plan (Section 2.3), as well as contracting downward flexibility (Section 2.2). On the other hand, load stages with higher demands (potential lower generation) are more suitable for signaling the VPL battery discharge/charge plan (Section 2.3), as well as contracting upward flexibility (Section 2.2).

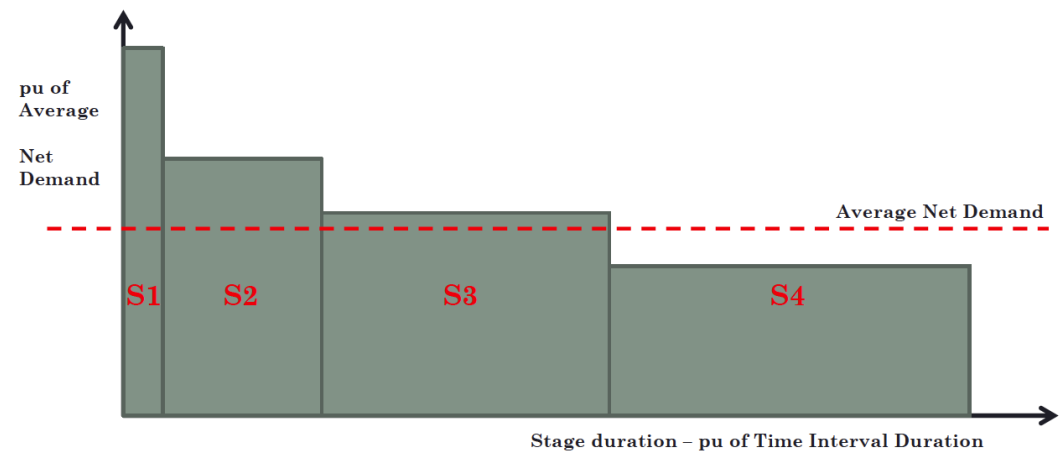


Figure 1. Load duration curve of net demand with 4 stages, S1, S2, S3 and S4.

2.2. Flexibility

In the context of the interaction between TSO, DSOs and VPPs, in the electricity sector, upward flexibility and downward flexibility, from the point of view of the TSO, refer to the ability of DSOs and VPPs to adjust their electricity demand in response to signals from TSO [55].

Upward flexibility, from the point of view of the TSO, refers to the ability of DSOs and VPPs to decrease their electricity demand, or increase the injection of distributed generation, when there is a shortage of electricity supply in the transmission network. This shortage of power can be generated by VRE sources connected to the grid, such as wind or solar power, which are subject to variability and intermittency, or due to generator outages or other factors. By reducing their demand during periods of a shortage of power supply, DSOs and VPPs can help to balance the electricity grid and prevent instability. The proposal of this paper models upward flexibility as a decrease in demand and VPP power injection, avoiding an increase in dispatchable generation.

Downward flexibility, on the other hand, refers to the ability of DSOs and VPPs to increase their electricity demand, or decrease the injection of distributed generation, when there is an excess of electricity supply in the transmission network. This can occur during periods of low demand or when there is an excess of power supply due to VRE sources connected to the grid. By increasing their demand during periods of excess of power supply, DSOs and VPPs can help to maintain grid stability and prevent curtailments and congestions. The proposal of this paper models downward flexibility as an increase in demand, avoiding renewable and non-dispatchable generation curtailment.

Upward and downward flexibility, considering power system operation, require sophisticated communication and coordination between TSO and DSOs, as well as advanced control and monitoring technologies. The use of digital platforms and real-time data analytics can help to facilitate this interaction and enable more efficient and reliable electricity grid operation.

2.3. Virtual Power Lines

The concept of virtual power lines (VPLs) is modeled in this paper considering energy storage systems (ESSs) that are used to defer the expansion of transmission lines by providing additional flexibility and control over the flow of electricity in the transmission system. Energy is stored during low-demand stages and discharged during high-demand stages. ESSs, used in this way, can better utilize existing transmission infrastructure and reduce or defer the need for new or expanded transmission lines [56,57]. This proposal models candidate virtual power lines as aggregated groups of ESSs, located in multiple transmission network nodes, with a charge and discharge status compatible with the high and low availability of renewable energy, avoiding generation curtailment due to transmission congestion.

In Figure 2, a diagram is presented where the operation of an VPL can be illustrated, with the aim of describing the VPL model used in this work.

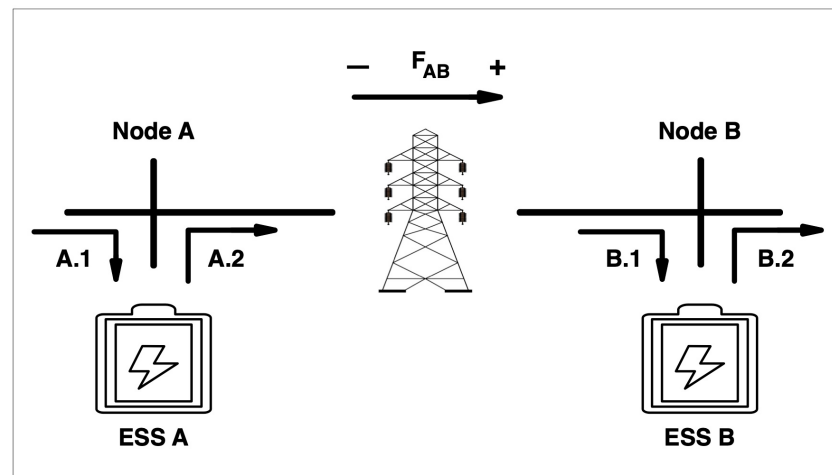


Figure 2. Virtual power line—steps.

In the nodes where each end of a physical transmission line that has implemented the VPL concept is connected (in this figure, called nodes A and node B), ESSs are implemented to accumulate and discharge energy at controlled times, (in this figure called ESS A and ESS B).

Each ESS unit operates permanently in a charging or discharging status. In the example, in Figure 2, the charge status of ESS A is indicated as A.1 and the discharge status as A.2, while the charge status of ESS B is indicated as B.1 and the discharge status as B.2.

The charging or discharging status of each ESS unit is controlled based on two indicators: the net demand stage, Figure 1, and the direction of power flow occurring on the transmission line.

To describe the four possible situations that an ESS unit can be in, depending on the net demand level and power flow in the transmission line, Table 2 presents the charging or discharging status of this unit for each situation.

The implementation of a real case in a transmission line was not part of the scope of this work.

Table 2. Virtual power line - battery wtatus.

	Demand Side Fij (+)	Supply Side Fij (−)
High grid usage	Discharge	Charge
Low grid usage	Charge	Discharge

2.4. Virtual Power Plants

Virtual power plants (VPPs) are defined as aggregated groups of distributed energy resources (DERs), represented as generators, ESSs and demands, located in multiple transmission network nodes, each one with a variable power output divided in stages with a similar concept as that described in Section 2.1 [58].

In this paper, for the proposed TGEP model, it is assumed that some parties' generation and associated demands are modeled as a set of VPPs that share the use of the transmissions system being planned.

In order to ensure the reliability and adequacy of the power system, dealing with an amount of non-dispatchable generation and uncertainties, the TGEP proposal considers the procurement and dispatch of a reserve capacity to handle unforeseen fluctuations in electricity supply or demand, which can arise due to various factors such as sudden changes in weather, equipment failures or unexpected spikes in electricity consumption. This reserve capacity is contracted from those parties modeled as VPPs. The planned capacity to be contracted from the reserve market area is considered as a planned cost, and at each stage that needs to be dispatched, the corresponding amount of energy cost is also computed.

2.5. Objective Function

The TGEP problem aims to minimize the total cost of investment and operation along the planning horizon. Considering the deterministic optimization, the objective function of the first implementation model is presented in (2). The objective function should minimize the investment cost of the expansion plan and the operating cost, considering the whole planning horizon, computing investment and operational cost to the present value.

$$\min \sum_t \frac{1}{(1+dr)^t} (C_{inv} + C_{opr} + C_{lc} + C_{vrec}) \quad (2)$$

where C_{inv} is the investment cost, C_{opr} is the operational and generation cost, C_{lc} is the load curtailment cost and C_{vrec} is the VRE curtailment cost.

To make investment projects with different useful life values comparable to each other, their useful life is extended to infinity, and the present value of the infinite series of installments is determined. This results in the sum of several installments, one in each period of time, brought to the present value at a previously defined discount rate. The perpetuity financial model procedure used is based in [59].

The investment cost C_{inv} is detailed in (3). It is related to investment in new line circuits, dispatchable and non-dispatchable generation and ESS units.

$$C_{inv} = \left[\begin{array}{l} \sum_{l,c} IC_{l,t}^{TL} Circ_{l,c,t} + \\ \sum_{cd,b} IC_{cd,b,t}^D ip_{cd,b,t}^{CD} + \\ \sum_{cnd,b,p} IC_{cnd,b,t}^{ND} ip_{cnd,b,t}^{CND} + \\ \sum_h IC_{h,t}^{ST} InST_{h,t} + \\ \sum_l IC_{l,t}^{VL} InVL_{l,t} \end{array} \right] \quad (3)$$

The operational cost C_{oper} is detailed in (4). It is related to the power provided by dispatchable generation, reserve generation (VPP), ESSs and congestion of transmission lines.

$$C_{oper} = \left[\begin{array}{l} \sum_{vp,b,t} OC_{vp,b,t}^{VPR} (p_{vp,b,t,w}^{VPRC} + q_{vp,b,t,w}^{VPRC}) + \\ \sum_{vp,b,t,s} OC_{vp,b,t,s}^{VPR} (p_{vp,b,t,s,w}^{VPR} + q_{vp,b,t,s,w}^{VPR}) + \\ \sum_{b,t,s} OC_{b,t,s}^D P_{b,t,s,w}^D + \sum_{b,t,s} OC_{b,t,s}^{CD} P_{b,t,s,w}^{CD} + \\ \sum_{b,t,s} OC_{b,t,s}^{FxU} P_{b,t,s,w}^{FxU} + \\ \sum_{b,t,s} OC_{b,t,s}^{FxD} P_{b,t,s,w}^{FxD} + \\ \sum_{b,t,s} OC_{b,t,s}^{dFxU} dP_{FxU}^{RSP} + \\ \sum_{b,t,s} OC_{b,t,s}^{dFxU} dP_{FxU}^{RSP} + \\ \sum_{h,t,s} OC_{h,t,s}^{ST} (p_{h,t,s,w}^{STC} + p_{h,t,s,w}^{STD}) \end{array} \right] \quad (4)$$

The load curtailment cost C_{lc} is detailed in (5). It is related to situations of a shortage of power or energy supply.

$$C_{lc} = \left[\sum_{b,s} OC_{b,t}^{LC} P_{b,t,s,w}^{LC} \right] \quad (5)$$

The VRE curtailment cost C_{vrec} is detailed in (6). It is related to situations of a shortage of demand or insufficient transmission capacity.

$$C_{vrec} = \sum_t \frac{1}{(1+dr)^t} \left[\sum_{b,s} OC_{b,t}^{NDC} P_{b,t,s,w}^{NDC} \right] \quad (6)$$

2.6. Power Balance Constraints

Power balance constraints are electrical properties that need to be guaranteed. They are used to establish the optimal dispatch in the power flow calculation.

$$p_{vp,b,t,s,w}^{VPR} + p_{vp,b,t,s,w}^{VPRC} + p_{vp,b,t,s,w}^{VPD} + p_{vp,b,t,s,w}^{VPND} + p_{b,t,s,w}^D + p_{cd,b,t,s,w}^{CD} + p_{cnd,b,t,s,w}^{CND} + p_{b,t,s,w}^{LC} + \sum_b f_{l,t,s,w} + \quad (7)$$

$$p_{b,t,s,w}^{FxU} + p_{h,t,s,w}^{STD} - p_{h,t,s,w}^{STC} + dP_{FxU}^{RSP} = p_{cnd,t,s,w}^{NDC} + ndP_{b,t,s,w} + dP_{FxU}^{RSP} + dP_{b,t,s}^{RSP} + p_{b,t,s,w}^{FxU} \quad \forall v, b, t, s$$

$$q_{vp,b,t,s,w}^{VPR} + q_{vp,b,t,s,w}^{VPD} + q_{vp,b,t,s,w}^{VPND} + q_{b,t,s,w}^D + q_{cd,b,t,s,w}^{CD} + q_{cnd,b,t,s,w}^{CND} + \sum_b f_{Ql,t,s,w} = ndQ_{b,t,s,w} + dQ_{b,t,s}^{RSP} \quad \forall v, b, t, s \quad (8)$$

2.7. Demand Response Constraints

These constraints are intended to guarantee the availability of energy supply and consumption services, within the limits contracted as the demand response.

$$dP_{FxU}^{RSP} \leq dP_{b,t,s}^{RSP} * dBand_{b,t,s}^{RSP} \quad \forall b, t, s \quad (9)$$

$$dP_{FxU}^{RSP} \leq dP_{b,t,s}^{RSP} * dBand_{b,t,s}^{RSP} \quad \forall b, t, s \quad (10)$$

2.8. Reference Bar and Voltage Constraints

These constraints are intended to ensure that the electrical variables associated with the voltage at the network nodes meet the electrical characteristic limits.

$$teta_{vref,t,s,w} = 0 \quad \forall t, s \quad (11)$$

$$V_{vref,t,s,w} = 1 \quad \forall t, s \quad (12)$$

$$\Delta V_{vref,t,s,w} = 0 \quad \forall t, s \quad (13)$$

$$v_{b,t,s,w} = 1 + \Delta V_{b,t,s,w} \quad \forall b, t, s \quad (14)$$

2.9. Transmission Line Circuits Constraints

These constraints are intended to guarantee the electrical characteristics of a transmission line.

$$\sum_c \text{Circ}_{l,c,t} \leq \text{Line_MaxCirc}_l \quad \forall l, t \quad (15)$$

$$|f_{l,t,s,w}| \leq \sum_c \text{LCirc_Cap} P_{l,c} \quad \forall l, t, s \quad (16)$$

$$f_{l,t,s,w} = (\Delta V_{b_{l_o},t,s,w} - \Delta V_{b_{l_d},t,s,w}) \text{admit} G_l - (\text{teta}_{b_{l_o},t,s,w} - \text{teta}_{b_{l_d},t,s,w}) \text{admit} B_l \quad \forall l, t, s \quad (17)$$

2.10. Transmission Line Circuits Constraints AC Linearized

These constraints are intended to guarantee the electrical characteristics of a transmission line, considering a linearized AC model.

$$fQ_{l,t,s,w} = - (1 + 2\Delta v_{b_{l_d},t,s,w}) \text{admit} B_{0l} - (\Delta v_{b_{l_o},t,s,w} - \Delta v_{b_{l_d},t,s,w}) \text{admit} B_l - (\text{teta}_{b_{l_o},t,s,w} - \text{teta}_{b_{l_d},t,s,w}) \text{admit} G_l \quad \forall l, t, s \quad (18)$$

2.11. Transmission Line Circuits Constraints AC—Second-Order Cone Constraint

These constraints are intended to guarantee the electrical characteristics of a transmission line, considering a linearized AC model, using a second-order cone constraint to consider apparent power constraint.

$$(f_{l,t,s,w})^2 + (fQ_{l,t,s,w})^2 \leq (\text{LCircCapVA}_l)^2 \quad \forall l, t, s \quad (19)$$

2.12. Energy Storage System Constraints

These constraints are intended to guarantee the electrical characteristics of battery energy storage power and energy limits.

$$ESS_{h,t,w} = \sum_s (p_{h,t,s,w}^{STC} - p_{h,t,s,w}^{STD}) pDur_s + ESS_{h,t-1,w} \quad \forall h, t \quad (20)$$

$$ESS_{h,t,w} \leq ESS_{h,t,w}^{CAP} \quad \forall h, t \quad (21)$$

$$p_{h,t,s,w}^{STC} \leq soc_{h,t,s,w} \cdot P_h^{STMax} \quad \forall h, t, s \quad (22)$$

$$p_{h,t,s,w}^{STD} \leq (1 - soc_{h,t,s,w}) \cdot P_h^{STMax} \quad \forall h, t, s \quad (23)$$

2.13. Virtual Power Line Constraints

These constraints are intended to ensure coordination of battery energy storage systems to implement VPL functionality.

$$f_{l,t,s,w}^{Sign1} = f_{l,t,s,w} \quad (24)$$

$$f_{l,t,s,w}^{Sign1} \leq VPL_Fij1_St_{l,t,p,s,w} * M \quad (25)$$

$$f_{l,t,s,w}^{Sign1} \geq (-M * (1 - VPL_Fij1_St_{l,t,p,s,w})) + (VPL_Fij1_St_{l,t,p,s,w} * 0.001) \quad (26)$$

$$VPL_Carg1_St_{l,t,s,w} = Pat_VPL_Ind_p + (1 - VPL_Fij1_St_{l,t,p}) * VPL_Fij2_St_{l,t,p} \quad (27)$$

$$VPL_Carg1_St_{l,t,s,w} + Pat_VPL_Ind_p \leq 1 + VPL_Fij1_St_{l,t,p} \quad (28)$$

$$f_{l,t,s,w}^{Sign2} = f_{l,t,s,w} \quad (29)$$

$$f_{l,t,s,w}^{Sign2} \leq VPL_Fij2_St_{l,t,p,s,w} * M \quad (30)$$

$$f_{l,t,s,w}^{Sign1} \geq (-M * (1 - VPL_Fij2_St_{l,t,p,s,w})) + (VPL_Fij1_St_{l,t,p,s,w} * 0.001) \quad (31)$$

$$VPL_Carg2_St_{l,t,s,w} = (1 - VPL_Carg1_St_{l,t,s,w}) \quad (32)$$

$$f_{l,t,s,w} \leq M \quad (33)$$

$$f_{l,t,s,w} \geq -M \quad (34)$$

2.14. Flexibility Constraints

These constraints are intended to guarantee the availability of energy supply and consumption services, within the limits contracted as flexibility.

$$P_{b,t,s}^{FxD} \leq (iFxD_{b,t}) * p_max_b^{FxD} \quad \forall b, t, s \quad (35)$$

$$P_{b,t,s}^{FxU} \leq (iFxU_{b,t}) * p_max_b^{FxU} \quad \forall b, t, s \quad (36)$$

2.15. Virtual Power Plants Constraints

These constraints are intended to guarantee the limits of power of VPP contracted generators.

$$p_{vp,b,t,s,w}^{VPD} \leq p_max_{vp}^{VPD} \quad \forall vp, b, t, s \quad (37)$$

$$q_{vp,b,t,s,w}^{VPD} \leq q_max_{vp}^{VPD} \quad \forall vp, b, t, s \quad (38)$$

$$p_{vp,b,t,s,w}^{VPR} \leq p_max_{vp}^{VPR} \quad \forall vp, b, t, s \quad (39)$$

$$q_{vp,b,t,s,w}^{VPR} \leq q_max_{vp}^{VPR} \quad \forall vp, b, t, s \quad (40)$$

2.16. Power Limits Constraints

These constraints are intended to guarantee the limits of power of general contracted generators.

$$p_{b,t,s,w}^D \leq p_max_b^D \quad \forall b, t, s \quad (41)$$

$$q_{b,t,s,w}^D \leq q_max_b^D \quad \forall b, t, s \quad (42)$$

$$p_{b,t,s,w}^{NDC} \leq P_{cnd,b,t,s,w}^{CND} \quad \forall b, t, s \quad (43)$$

$$p_{b,t,s,w}^{CD} \leq p_max_b^{CD} \quad \forall b, t, s \quad (44)$$

$$q_{b,t,s,w}^{CD} \leq q_max_b^{CD} \quad \forall b, t, s \quad (45)$$

3. Problem Formulation—Modeling Uncertainties

The data-driven distributionally robust optimization (DDDRO) method has been developed to identify the worst-case probability distribution across a range of ambiguities, effectively integrating elements of stochastic programming (SP) and robust optimization (RO) [10]. This approach utilizes historical data to craft various scenarios, applying worst-case probabilities backed by a moment-based ambiguity set to establish the probability distribution. DDDRO allows for the formulation of a two-stage robust optimization problem, aiming to determine the maximum cost within an uncertainty set. As the number of scenarios grows and the problem becomes nonlinear, the complexity rises significantly. To address this, a duality-free decomposition method is used, in order to transform the bi-level (max-min) problem into two independent subproblems [60,61].

Using DDDRO to model uncertainties, the objective function presented in (2) is reorganized. The variables associated with investments are considered decision variables in the first level. In this level, the decision variables that minimize the investment costs of candidate transmission lines, dispatchable generation, planned VRE generation and VPL BESS infrastructure are optimized ($IC_{l,t}^{TL}$, $IC_{cd,b,t}^D$, $IC_{cnd,b,t}^{ND}$, $IC_{h,t}^{ST}$, $IC_{l,t}^{VL}$).

At the second level, the variables' related uncertainties, net demand and candidate VRE generation are optimized ($p_{cnd,b,t,s,w}^{CND}$, $ndP_{b,t,s,w}$, $ndQ_{b,t,s,w}$), considering the maximization, worst realization and high operational cost.

At the third level, the objective function presented in (2) is minimized, with the determination of the values of the vector of operating variables ($p_{v,b,t,s,w}^{VPR}$, $q_{v,b,t,s,w}^{VPR}$, $p_{v,b,t,s,w}^{VPD}$, $q_{v,b,t,s,w}^{VPD}$, $p_{v,b,t,s,w}^{VPND}$, $q_{v,b,t,s,w}^{VPND}$, $p_{b,t,s,w}^D$, $q_{b,t,s,w}^D$, $p_{b,t,s,w}^{CD}$, $q_{b,t,s,w}^{CD}$, $p_{b,t,s,w}^{ND}$, $q_{b,t,s,w}^{ND}$, $p_{b,t,s,w}^{CND}$, $q_{b,t,s,w}^{CND}$, $p_{b,t,s,w}^{FxD}$, $q_{b,t,s,w}^{FxD}$, $iFxD_{b,s,w}$, $iFxD_{b,s,w}$, $dP_{b,t,s,w}$, $dQ_{b,t,s,w}$, $dP_{b,t,s,w}^{RSP}$, $dQ_{b,t,s,w}^{RSP}$, $p_{b,t,s,w}^{STD}$, $q_{b,t,s,w}^{STD}$, $p_{h,t,s,w}^{STC}$, $q_{h,t,s,w}^{STC}$, $v_{b,t,s,w}$, $\theta_{ij,t,s,w}$).

Details of this modeling can be found at [18].

This three level model, as presented in (60), is solved using column and constraint generation as proposed in [62]. The duality-free decomposition method is used to transform the bi-level (max-min) problem into independent subproblems.

4. Solution Procedure

4.1. Deterministic Procedure

TGEP problems are considered computationally heavy to solve. These planning problems involve a large number of integer and continuous variables and constraints, especially when considering long-term planning horizons and multiple scenarios or uncertainties. The complexity increases with the size of the power system, the number of potential new generation units and the possible expansion options for the transmission network. Solving large mixed-integer linear or nonlinear optimization problems requires sophisticated numerical algorithms and optimization techniques. As the size of the problem grows, the search space becomes larger, and it becomes more challenging to find the optimal solution within a reasonable time frame.

One approach to deal with this complexity is distributed optimization. ADMM is proposed in [63,64], and the ability to achieve a converged solution depends on the penalty parameter tuning. Another approach is the use of decomposition; the main methods that have been used are Benders [65] and constraint and column generation (CCG), which is considered to converge faster than Benders decomposition [66,67].

The proposed expansion model will be solved using CCG decomposition as defined in [62]:

$$\min_y (c^T y + \eta) \quad (46)$$

s.t.

$$Ay \geq d \quad (47)$$

$$\eta \geq b^T x^l, \quad l = 1, \dots, r \quad (48)$$

$$Ey + G x^l \geq h, \quad l = 1, \dots, r \quad (49)$$

$$y \in S_y \quad (50)$$

$$x^l \in S_x, \quad l = 1, \dots, r \quad (51)$$

$$S_y \subseteq \mathbb{R}^n, \quad S_x \subseteq \mathbb{R}^m \quad (52)$$

The decision variables of vector y of (50) are the binary variables referring to investments in transmission lines, storage, generation units and virtual power lines. These are the first-stage variables of CCG decomposition. Vector c of (53) defines the investment costs.

The decision variables of vector x^l of (48) are the recourse decision variables of second-stage of CCG decomposition, related to the operating conditions of the planned power system at each stage. These second-stage variables $[x^1, x^2, \dots, x^r]$ represent the CCG columns that are created at each solution procedure step.

Solution procedure algorithm (CCG):

1. Set $LB = -\infty$, $UB = +\infty$, $k = 0$ and $O = \emptyset$
2. Solve the following master problem:

$$\min_{y, \eta} (c^T y + \eta) \quad (53)$$

s.t.

$$Ay \geq d \quad (54)$$

$$\eta \geq b^T x^l, \quad \forall l \in O \quad (55)$$

$$Ey + G x^l \geq h, \quad \forall l \leq k \quad (56)$$

Solution: $(y_{k+1}^*, \eta_{k+1}^*, x^{1*}, \dots, x^{k*})$

3. Update $LB = (c^T y_{k+1}^* + \eta_{k+1}^*)$
4. Solve the following slave problem:

$$\min_x (b^T x) \quad (57)$$

s.t.

$$G x^l \geq h - Ey_{k+1}^*, \quad \forall l \leq k \quad (58)$$

5. Update $UB = \min [UB, (c^T y_{k+1}^*)]$
6. If $(UB - LB) \leq \varepsilon$ return (y_{k+1}^*) and finish
7. Create variables (x^{k+1})
8. Add the following constraints to the master problem:

$$Ey + G x^{k+1} \geq h \quad (59)$$

9. Update $k = k + 1$ and go to Step 2

Figure 3 presents a detailed flowchart outlining the computational optimization procedure used to develop the proposed expansion plan. This systematic process is broken down into key activities, each enclosed in boxes labeled from M01 to M09, facilitating clear

reference and understanding. These steps are integral to the overall methodology, ensuring a structured and efficient approach to achieving the specified expansion plan.

Activities M01 to M03 are related to the input processes of all model configuration parameters and data related to the networks whose expansion plan will be determined.

Activities M04 to M06 are related to the generation of all time series that will be provided as input for the optimization activities. The generation of load duration curves, clustering and scenario generation, described in Section 5.1, are performed in these activities.

Activity M07 is related to a linear integer programming optimization where binary variables related to investments are determined.

Activity M08 is related to a linear programming optimization where operation optimization variables, considering uncertainty scenarios, are processed. These activities are presented in more detail in Section 4.2.

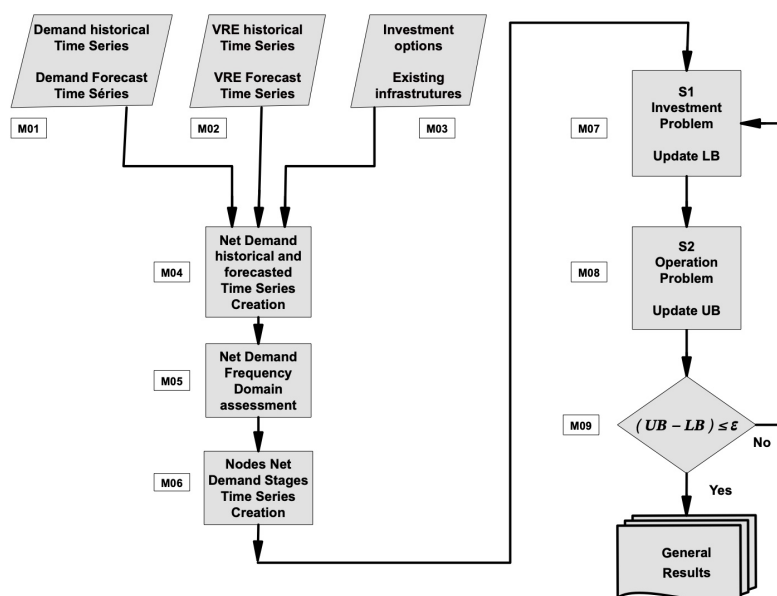


Figure 3. Expansion planning proposed methodology.

4.2. Procedure Considering Uncertainties

In order to consider uncertainties related to net demand and candidate VRE, module M08 of Figure 3 is decomposed in two problems. An upper level determines the maximum cost within an uncertainty set. The lower level minimizes the operational costs of each scenario corresponding to an uncertainty set.

$$\max_{w \in \Omega^W} \pi_w^0 \left(\min_{x \in \Omega(y, w)} b^t x \right) \quad (60)$$

4.2.1. Ambiguity Set

Probability distribution functions related to uncertain variables may not be available. As an alternative, using historical data is an option to obtain an approximation of the probabilities of a scenario of interest. Historical data can be converted into data bins, where an estimated probability distribution function (E-PDF) is established from the data bins, which allows the definition of the true probability distribution function (T-PDF) within a tolerance range. In [10], a confidence uncertainty set is proposed to cover all possible probability realizations by making the most of historical data and then estimating the distribution of worst-case uncertainties for all scenarios (Ω^W) according to the number of data bins (MD). Two norms L_1 and L_∞ are used to construct the confidence uncertainty set

based on historical data, which has been proved to converge to truth probability distribution when data points increase to infinity [68]. The confidence uncertainty set can be formulated as (61) [10].

When N historical data points are partitioned into M_D bins, data points in each sample bin, sequentially denoted as $N_1, N_2, \dots, N_D$, and the E-PDF can be estimated as $\pi_1^0 = N_1/N, \dots, \pi_D^0 = N_D/N$.

$$\Psi^{amb} = \left\{ \pi_w \left| \begin{array}{l} \|\pi_w - \pi_w^0\|_1 = \sum_{w=1}^{M_D} |\pi_w - \pi_w^0| \leq \delta_1 \\ \|\pi_w - \pi_w^0\|_\infty = \max_{1 \leq w \leq M_D} |\pi_w - \pi_w^0| \leq \delta_\infty \\ \sum_{w=1}^{M_D} \pi_w = 1 \\ \pi_w \geq 0, w = 1, 2, \dots, M_D \end{array} \right. \right\} \quad (61)$$

The right-hand thresholds δ_1 and δ_∞ in (61) refer to the tolerance coefficients associated with the given confidence level and historical data. As more historical data are inserted, the uncertainty set shrinks and E-PDF gets closer to the T-PDF. If the confidence levels of two norms are described as α_1 and α_2 , the tolerance coefficients can be reformulated as (62) [69].

$$\delta_1 = \frac{M_D}{2N} \ln\left(\frac{2M_D}{1 - \varphi_1}\right); \delta_\infty = \frac{1}{2N} \ln\left(\frac{2M_D}{1 - \varphi_\infty}\right) \quad (62)$$

4.2.2. Duality-Free Approach

The decomposition module M08 of Figure 3 in two problems (60) is necessary to find the critical scenario of the uncertainty set that provides an upper bound (UB). New variables and constraints are added to the master problem M07 to obtain a lower bound (LB). The master-problem and the decomposed M08 are solved iteratively (ℓ_{th}) and the method stops until the relative difference between the upper and lower bounds is less than a preset convergence tolerance $\mathcal{E}$.

The subproblem is a max-min bi-level problem with a structure that can be decomposed into several small subproblems without the duality information [60]. Given that between constraints associated with $\mathbf{x}$ and those associated with $\mathbf{w}$, there are no variables in common, the feasible region bounded by the variables $\mathbf{x}$ is disjointed with the ambiguity set Ψ^{amb} .

For each scenario w , an optimal solution x is obtained through an internal minimization problem, and this solution is fixed to an external maximization problem, to find the probability of the worst-case scenario w .

5. Case Studies

This section considers two case studies to validate the proposed model to optimize the transmission and generation expansion planning process, considering virtual power lines, virtual power plants, distributed energy injection and demand response flexibility. The implementation was programmed in Python 3.11.0, using Spyder 5.4.3, Pyomo 6.5.0 and Gurobi 10. All processing was performed using an Apple Studio M1 64 GB. The data for historical demand and VRE generation were obtained from [70], considering the period from 01/2015 to 12/2023 related to Spain, converted to p.u. in order to be used with the two systems in Sections 5.3 and 5.4. Files with data relating to investment, operating costs and demands used in this research can be obtained at [71].

For the case studies, information from two transmission networks with data and expansion planning results that can be found in the literature was used. The Garver and IEEE RTS-GMLC test system networks were used. The first, GARVER, was chosen to play the role of a small system, where the VPL and flexibility models proposed in this paper could be presented and tested in a simpler way. The second, IEEE RTS-GMLC, was chosen

to play the role of a medium/large system, where scalability and feasibility issues could be validated.

Initially, the model proposed in this article was used in two scenarios, Sections 5.3.1 and 5.4.1, considered base cases, for contrast or comparison with existing models used in the literature that is reviewed. These cases consider an expansion plan using models in common with the compared proposals, such as transmission lines, generators and loads. The first scenario uses the GARVER network and the second uses the IEEE RTS-GMLC test system with results found in the literature in references [72] and [73], respectively.

5.1. Cluster of Data Bins

Data-driven methodology can produce typical demand scenarios and related confidence uncertainty sets. Hourly historical data on VRE injection and demand power are proposed and available at [73] covering the period of one year. To use the data-driven methodology with a greater amount of historical data, a period of nine years was considered, with 15-minute interval measurements, from [70], as described in Section 5. Considering the results reported in [74], the sample space was partitioned into six data bins to represent random output net demand and produce discrete probability distributions with good results.

Figure 4 presents an example of the historical data that was used, showing the historical series of average measurements for each hour of one of the historical years used.

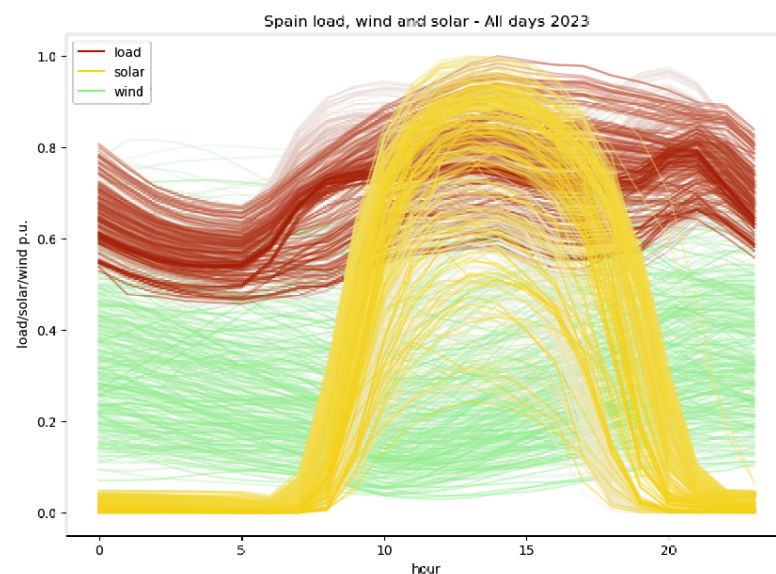


Figure 4. ENTSO-E one year data summarized.

Considering this historical time series, related to Spain 2015–2023 data available on [70], clustering methods such as K-means were employed to aggregate all data points into a representative scenario in each bin so that the estimated probability distribution is obtained by counting the frequency of data samples falling into each bin, considering the averaged metric of demand and VRE injection (wind and solar) as representative scenarios. Considering this context, Figure 5 shows six clusters (scenarios) as a function of the time of a day (15 min period).

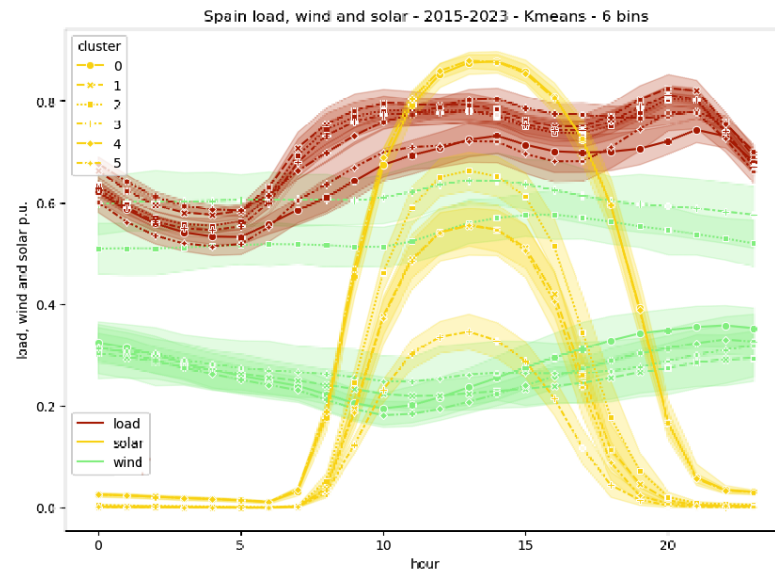


Figure 5. Clustering with six data BINs using K-means.

Figure 6 shows the probabilities of each cluster (scenario) obtained from the clustering process.

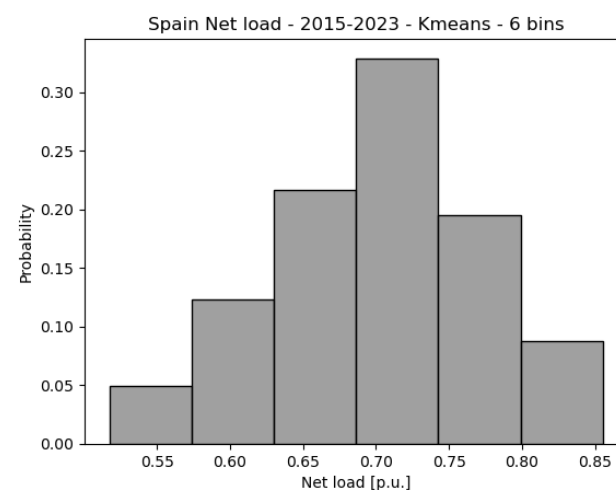


Figure 6. Probabilities of each cluster (scenario).

In this proposal, net demands are modeled using the load duration curve, which shows the relationship between the accumulated load and the percentage of time in which this load occurs. The planning model uses net demand for a given time interval divided into steps, which are related to the net load block curves, which divide the expected net load levels into discrete time blocks. Each time block of a considered net load demand block is called a stage.

Considering each of the six scenarios of demand, and those of VRE injection, which are all linked in time, a procedure was carried out to generate six scenarios, still linked in time, of net demand. These net demand time series were then processed and converted to typical day stages.

Each of the stages has a duration and a value, a proportion (p.u) in relation to the average net load of the day. The sum of the product between durations and values is equal to 1, in order to keep the average daily net demand unchanged. In these case studies, four net demand stages per day were considered, with the stage corresponding to peak moments. Stage number 1 was considered to have 0.05/1 of the duration. The next two stages each

had 0.2/1 of the duration. The last stage, corresponding to low-demand moments, stage number 4, was considered to have the remaining 0.55/1 of the duration.

Each of the stages has a duration and a value, a proportion (p.u) in relation to the average net load of the day. The sum of the product between durations and values is equal to 1, in order to keep the average daily net demand unchanged. Considering the actual time series used, Figure 7 presents an example of actual values considering Spain's net demand profile.

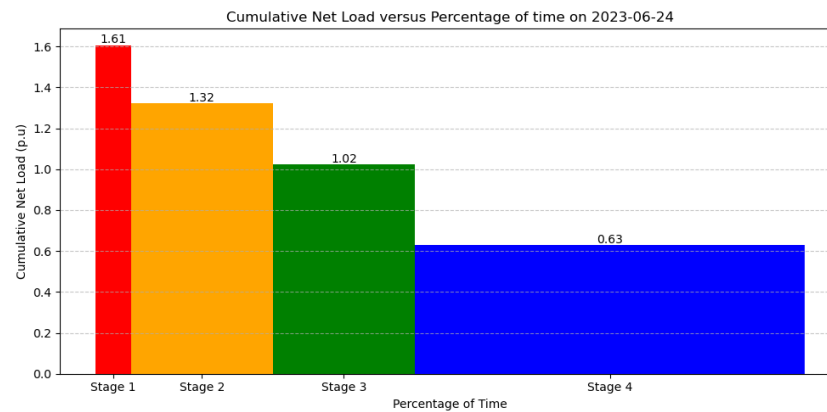


Figure 7. Spain data. Cumulative net load; typical day, discretized into four average net demand levels (S1, S2, S3, and S4)

5.2. Presentation of Case Studies Results

Each case study will be presented with a summary table with the relevant investment and cost components of the minimized objective function corresponding to the optimal expansion plan established for each case, computed to the present value.

For each case study, the result of the chosen expansion plan will be shown, presenting in a table a summary of the financial values, computed to the present value, of the investments and operating costs chosen to present for analysis and comparisons. This is a summarized view of the chosen expansion plan.

The variables that are used in the optimization procedure form a vector of network operation variables presented in Nomenclature, vector notation. These variables participate in operational constraints related to the electrical system, equipment specifications and resource limits. Almost all operation variables participate in the evaluation of the objective function that is minimized by the optimization procedure to choose the optimal expansion plan.

The summary presentation table has six columns with the information of new circuits installed, circuit of installed VPL, total active power of dispatchable generators, total active power of non dispatchable generators, total active power provided or consumed by demand response service and total active power provided or consumed by flexibility services. The last column of the table, the cost column, presents the total cost of the corresponding line of the table, computed to the present value, that was considered during the objective function computation of the optimal expansion plan presented by the table.

5.3. Garver 6-Node Network

The Garver 6-node network consists of 15 right-of-ways, one isolated node, a total load of 760 MW and 152 MVar and a total active power generation of 1140 MW. The complete data of this network can be obtained from [72].

The candidate ESSs considered are battery storage devices with a maximum charge and discharge rate of 50 MW, a round-trip efficiency of 85 % and a usable energy storage capacity of up to 75 MWh. For the investment values and operational costs related to

the ESS candidates for VPL, data from case studies and projections presented in [75,76] were used.

For this first case study, three scenarios were evaluated: scenario S1.1, considering only dispatchable generation, transmission data and demands obtained from [72]; scenario S1.2, considering candidate VPL using actual ESS investment and operation costs obtained from [75,76]; and scenario S1.3, considering a projected reduction in ESS investment and operation costs based in [75].

5.3.1. Garver 6-Node Network—Scenario S1.1

Scenario S1.1 has the objective to be a base case for comparisons. It considers only dispatchable generation, existing and candidate. Transmission lines data, and demands are obtained from [72].

Table 3 presents the results of the expansion planning for scenario S1.1, which will be used as a base case for comparisons with other scenarios.

The summary of the results of the optimal expansion plan shows that four transmission line circuits are planned. The operation cost related to generation is related only to dispatchable generation.

Considering that this scenario was created for contrast or comparison with other models in the reviewed literature, it can be observed that the expansion plan obtained is compatible with the solution found in [72].

Table 3. Scenario S1.1 Results.

Scenario S1.1						
New Circuits	VPL	Disp Gen [GW]	Ndisp Gen [GW]	Dem Response [GW]	DSO Flex [GW]	Cost [M\$]
2–6	-	-	-	-	-	30
2–6	-	-	-	-	-	30
3–5	-	-	-	-	-	20
4–6	-	-	-	-	-	30
-	-	0.68	-	-	-	28.2
Total						138.2

5.3.2. Garver 6-Node Network—Scenario S1.2

Scenario S1.2 aims to identify the impact of using VPL on investments in the expansion of transmission infrastructure, either replacing investments or delaying investments in transmission. It is considered the candidate VPL that it is possible to deploy, using real ESS investment and operation costs obtained from [75,76].

Table 4 presents the results of scenario S1.2 expansion planning.

Table 4. Scenario S1.2 Results.

Scenario S1.2						
New Circuits	VPL	Disp Gen [GW]	Ndisp Gen [GW]	Dem Response [GW]	DSO Flex [GW]	Cost [M\$]
2–6	-	-	-	-	-	30
-	2–6	-	-	-	-	27.5
3–5	-	-	-	-	-	20
-	4–6	-	-	-	-	27.5
-	-	0.68	-	-	-	28.2
Total						133.2

The summary of the results of the optimal expansion plan related to Scenario S1.2, when compared with the results of Scenario S1.1, shows that only two of the previous transmission line circuits are planned as investments; circuits 2–6 and 4–6 are replaced by

less expensive VPL infrastructure. The operation cost related to generation is related only to dispatchable generation.

5.3.3. Garver 6-Node Network—Scenario S1.3

Scenario 1.3 has the objective of making a sensitivity analysis of the impact of ESS cost reduction and its use as transmission investment reduction or postponement. It is considered a projected reduction in ESS investment and operation costs based in [75].

Table 5 presents the results of scenario S1.3 expansion planning.

The summary of the results of the optimal expansion plan related to Scenario S1.3, when compared with the results of Scenario S1.2, shows that only one of the previous transmission line circuits is planned as investment; circuits 2–6, 3–5 and 4–6 are replaced by less expensive VPL infrastructure. The operation cost related to generation is related only to dispatchable generation.

Table 5. Scenario S1.3 Results.

Scenario S1.3						
New Circuits	VPL	Disp Gen [GW]	Ndisp Gen [GW]	Dem Response [GW]	DSO Flex [GW]	Cost [M\$]
2–6	-	-	-	-	-	30
-	2–6	-	-	-	-	18.75
-	3–5	-	-	-	-	18.75
-	4–6	-	-	-	-	18.75
-	-	0.68	-	-	-	28.2
Total						114.45

5.4. IEEE RTS-GMLC

The IEEE RTS-GMLC test system consists of 104 right-of-ways, 36 at 138 kV and 68 at 230 kV, and 16 power transformers. The RTS-GMLC proposes time series considering VRE injection and demand; it considers one hour for day-ahead data and 15 minutes for real-time data. The complete data of this network can be obtained from [73].

The candidate ESSs considered are battery storage devices with a maximum charge and discharge rate of 50 MW, a round-trip efficiency of 85 % and a usable energy storage capacity of up to 75 MWh.

For this second case study, four scenarios were evaluated: scenario S2.1, considering only dispatchable generation, transmission data and demands obtained from [73]; scenario S2.2, considering candidate VPL using actual ESS investment and operation costs obtained from [75,76]; scenario S2.3, considering a projected reduction in ESS investment and operation costs based in [75]; and scenario S2.4, considering renewable generation investment, demand response and flexibility acquired from TSO-DSO interconnection. The model was parameterized to execute a three-year expansion plan, considering an annual linear growth rate of 4.5% for both demand and VRE.

Considering the computational effort, the processing time to obtain the solution was approximately 10 minutes, on average, for the scenarios considered in this case. The scenarios with the largest number of integer variables to be optimized, related to lines, VPL, storage and generation candidates for installation, reached a processing time of around 30 minutes. These values obtained show that the scalability of computational effort is adequate for use in medium or large-scale problems. The IEEE RTS test system is divided into three generation dispatch areas that can be used to distribute the execution of the optimization model. Without using this resource, adequate performance was achieved.

5.4.1. IEEE RTS-GMLC—Scenario S2.1

Scenario S2.1 has the objective of being a base case for comparisons. It considers only dispatchable generation, existing and candidate. Transmission lines data and demands are obtained from [73].

Table 6 presents the results of scenario S2.1 expansion planning.

Considering that this scenario was created for contrast or comparison with other models in the reviewed literature, it can be observed that the expansion plan obtained is compatible with the solutions found in [73].

There is no operational cost related to non-dispatchable generation, demand response and contracted flexibility, since only dispatchable generation is considered in this scenario.

Table 6. Scenario S2.1 Results.

Scenario S2.1						
New Circuits	VPL	Disp Gen [GW]	Ndisp Gen [GW]	Dem Response [GW]	DSO Flex [GW]	Cost [M\$]
15–24	-	-	-	-	-	99.8
55–56	-	-	-	-	-	39.3
59–61	-	-	-	-	-	24.9
58–60	-	-	-	-	-	14.7
-	-	8.2	-	-	-	348
Total						526.7

Considering that this scenario was created for contrast or comparison with other models in the reviewed literature, it can be observed that the expansion plan obtained is compatible with the solutions found in [73].

There is no operational cost related to non-dispatchable generation, demand response and contracted flexibility, since only dispatchable generation is considered in this scenario.

5.4.2. IEEE RTS-GMLC—Scenario S2.2

Scenario S2.2 aims to identify the impact of using VPL on investments in the expansion of transmission infrastructure, either replacing investments or delaying investments in transmission. It is considered a candidate VPL that it is possible to deploy, using real ESS investment and operation costs obtained from [75,76].

Table 7 presents the results of scenario S2.2 expansion planning.

Table 7. Scenario S2.2 Results.

Scenario S2.2						
New Circuits	VPL	Disp Gen [GW]	Ndisp Gen [GW]	Dem Response [GW]	DSO Flex [GW]	Cost [M\$]
-	15–24	-	-	-	-	37.5
-	55–56	-	-	-	-	37.5
59–61	-	-	-	-	-	24.9
58–60	-	-	-	-	-	14.7
-	-	8.2	-	-	-	348
Total						462.6

There is no operational cost related to non-dispatchable generation, demand response and contracted flexibility, since only dispatchable generation is considered in this scenario.

Figure 8 presents detailed information about the times when transmission line circuits are planned to be deployed.

The cost reductions observed when integrating VPNs compared to investments in conventional transmission can be justified mainly by the fact that, by operating ESSs in a coordinated manner, the peak usage time intervals of a transmission line can be reduced

by shifting this energy traffic to times of low usage. This is a procedure equivalent to the peak shaving of load curves. Thus, having a lower peak usage, for a transmission line, with the growth in demand that is planned to occur, its expansion may not be necessary or be delayed in time.

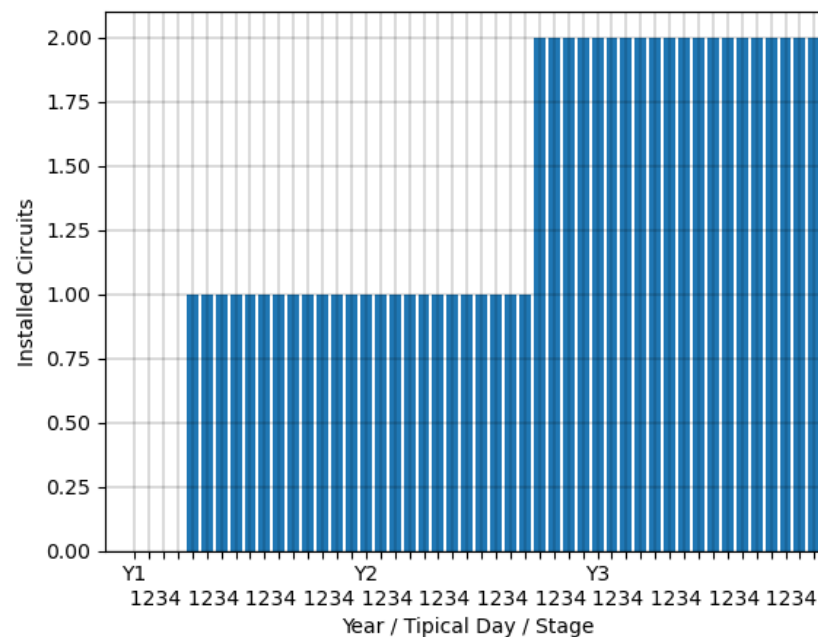


Figure 8. Transmission line circuits deployment time.

5.4.3. IEEE RTS-GMLC—Scenario S2.3

Scenario 2.3 has the objective of making a sensitivity analysis of the impact of ESS cost decline and its use as transmission investment reduction or postponement. It is considered a projected reduction in ESS investment and operation costs based in [75].

Table 8 presents the results of scenario S2.3 expansion planning.

Table 8. Scenario S2.3 Results.

Scenario S2.3						
New Circuits	VPL	Disp Gen [GW]	Ndisp Gen [GW]	Dem Response [GW]	DSO Flex [GW]	Cost [M\$]
-	15–24	-	-	-	-	22.5
-	55–56	-	-	-	-	22.5
-	59–61	-	-	-	-	22.5
58–60	-	-	-	-	-	14.7
-	-	8.2	-	-	-	338
Total						420.2

There is no operational cost related to non-dispatchable generation, demand response and contracted flexibility, since only dispatchable generation is considered in this scenario.

5.4.4. IEEE RTS-GMLC—Scenario S2.4

Scenario 2.4 has the objective of considering the impact of an available option to use renewable generation investment, with demand response and flexibility acquired from TSO-DSO interconnection. The model was parameterized to execute a three-year expansion plan, considering an annual linear growth rate of 4.5% for both demand and VRE. It is considered that 25% of the demand has contracts to provide a demand response service, and the upward and downward flexibility contracted is 30%.

Table 9 presents the results of scenario S2.4 expansion planning.

Table 10. Line usage and congestion indicator.

Scenario	Average Line Usage [p.u.]	Line Losses [p.u. h]	LMP Average [\$]
S2.1	0.968	450.3	1001.3
S2.2	1.167	447.4	981.1
S2.3	1.185	443.0	731.5
S2.4	1.196	440.5	66.8

6. Conclusions

This article presents a new model for planning the expansion of a transmission and generation system considering the impact of virtual power lines, with the investment in energy storage in the transmission system, in order to have a reduction or a postponement of investments in transmission lines, as well improving the electrical power system adequacy as a whole. The ability to contract flexibility from the TSO-DSO interconnection is modeled, in order to consider a reduction in system expansion investments, considering both energy and capacity reserve. The formulation developed includes a linear AC-OPF model with the incorporation of reactive power modeling in the GTEP problem. In order to make the resulting problem tractable when formulated for medium-to-large-scale systems, the daily time step used when computing the system operation uses the concept of a variable time net demand stage, obtained using the load duration curve, that shows the relationship between the cumulative load and the percentage of time for which that load occurs.

Flexibility is provided to the AC power flow transmission network model by distribution systems providing upward and downward flexibility services, related to demand response, distributed generation or energy storage systems, aggregated as virtual power plants.

A data-driven distributionally robust optimization-DDDRO approach is proposed to consider uncertainties of demand and variable renewable energy generation.

A novel optimization model for expanding generation and transmission infrastructure is introduced, using original modeling and the application of battery energy storage to operate in coordination with the load profile of transmission lines. It is used to consider storage capacity as an alternative to investing in the expansion of traditional transmission lines. The proposed model also considers the use of variable renewable energy to meet demands, and aggregated energy, power capacity and flexibility, provided to the transmission operation of the power system.

The proposed formulation has been applied to compute the GTEP for the RTS-GMLC test system that is a medium-sized system. This allowed evaluating the performance and computational time compared to calculating expansion plans determined with DC-OPF formulations.

The results obtained demonstrate an approximate 15% reduction in the total costs of the solution obtained in the test systems used. It is also demonstrated that there is an approximate 20% improvement in the efficient use of the transmission system. Finally, an improvement in the locational marginal pricing indicator of the transmission system is also evident. The proposed model supports the consideration and deployment of technologies and services to support the design of modern power systems to support the needs of a low-carbon emission context.

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Nomenclature

Sets

Ω^D	Dispatchable generation units
Ω^{CD}	Candidate dispatchable generation units
Ω^{ND}	Non-dispatchable generation units
Ω^{CND}	Candidate non-dispatchable generation units
Ω^{VPD}	VPP dispatchable generation units
Ω^{VPND}	VPP non-dispatchable generation units
Ω^H	Battery storage units
Ω^{VL}	Virtual power lines
Ω^{VP}	Virtual power plants
Ω^S	Demand stages
Ω^{CS}	Candidate storage units
Ω^B	Set of nodes in the power transmission network
Ω^L	Set of lines in the power transmission network, $\Omega^L \subseteq \Omega^B \times \Omega^B$
Ω^C	Set of circuits in the power transmission network line
Ω^W	Set of scenarios
Ψ^{amb}	DDDRO ambiguity set

Indices

b	Node, $b \in \Omega^B$
c	Line circuit, $c \in \Omega^C$
cd	Candidate dispatchable generation, $cd \in \Omega^{CD}$
cnd	Candidate non-dispatchable generation, $cnd \in \Omega^{CND}$
h	Battery storage unit, $h \in \Omega^H$
l	Line, $l \in \Omega^L$
s	Stage, $s \in \Omega^S$
t	Time step, t
vl	Virtual power line, $vl \in \Omega^{VL}$
vp	Virtual power plant, $vp \in \Omega^{VP}$
w	Scenario, $w \in \Omega^W$

Input Data and Operators

$\mathcal{T}$	Time horizon of the problem
a	Area
b_l	Susceptance [p.u.] of line l
dr	Discount rate
g_l	Conductance (p.u.) of line l
$Line_MaxCirc_l$	Maximum number of circuits of line l
$LCirc_CapP_{l,c}$	Active power capacity of circuit c of line l
$LCirc_CapVA_{l,c}$	Apparent power capacity of circuit c of line l
$IC_{l,t}^{TL}$	Investment cost of additional line circuit at corridor l in time period t [\$/circuit]
$IC_{cd,b,t}^D$	Investment cost of additional dispatchable generation at node b in time period t [\$/MW]
$IC_{cnd,b,t}^{ND}$	Investment cost of additional non-dispatchable generation at node b in time period t [\$/MW]
$IC_{h,t}^{ST}$	Investment cost of battery storage h in time period t [\$/MW h]
$IC_{vl,t}^{VL}$	Investment cost of VPL vl in time period t [\$/MW h]
$OC_{b,t}^D$	Variable cost of existing dispatchable generation at node b in time

	period t [\$/MW h]
$OC_{b,t}^{CD}$	Variable cost of candidate dispatchable generation at node b in time period t [\$/MW h]
$OC_{b,t}^{FxU}$	Variable cost of upward flexibility at node b in time period t [\$/MW h]
$OC_{b,t}^{FxD}$	Variable cost of downward flexibility at node b in time period t [\$/MW h]
$OC_{b,t}^{dFxU}$	Variable cost of demand response upward flexibility at node b in time period t [\$/MW h]
$OC_{b,t}^{dFxD}$	Variable cost of demand response downward flexibility at node b in time period t [\$/MW h]
$OC_{h,t}^{ST}$	Variable cost of storage h in time period t [\$/MW h]
$OC_{b,t}^{VPR}$	Variable cost of P2P active power contracted at node b in time period t [\$/MVA]
$OC_{b,t}^{VPRG}$	Variable cost of P2P active generation contracted at node b in time period t [\$/MW h]
$OC_{b,t}^{LC}$	Variable cost of load curtailment at node b in time period t [\$/MW h]
$OC_{b,t}^{NDC}$	Variable cost of non-dispatchable generation curtailment at node b in time period t [\$/MW h]
OC_t^{Cong}	Variable cost of congestion in time period t [\$/MW h]
π_w	Probability of scenario w
π_w^0	Probability of scenario w from data
$dP_{b,t,s}^{RSP}$	Active power of demand response, bus b , time period t , demand stage s [MW]
$dQ_{b,t,s}^{RSP}$	Reactive power of demand response, bus b , time period t , demand stage s [MVar]
$dBand_{b,t,s}^{RSP}$	Demand response available flexibility band, bus b , time period t , demand stage s [$p.u.$]
$ndP_{b,t,s,w}$	Active power of net demand, bus b , time period t , demand stage s , scenario w [MW]
$ndQ_{b,t,s,w}$	Reactive power of net demand, bus b , time period t , demand stage s , scenario w [MVar]
$p_{b,t,s,w}^{CND}$	Active power of candidate non-dispatchable generation units, bus b , time period t , demand stage s , scenario w [MW]
$q_{b,t,s,w}^{CND}$	Reactive power of candidate non-dispatchable generation units, bus b , time period t , demand stage s , scenario w [MVar]
$p_{vp,b,t}^{VPRC}$	Active power of VPP contracted in the P2P market, vpp vp , bus b , time period t [MW]
$q_{vp,b,t}^{VPRC}$	Reactive power of VPP contracted in the P2P market, vpp vp , bus b , time period t [MVar]
$avlD_{b,t}$	Active power available as downward flexibility at bus b , time period t [MW]
$avlU_{b,t}$	Active power available as upward flexibility at bus b , time period t [MW]
$ESS_{h,t}^{CAP}$	Energy capacity of battery storage h , time period t [MW h]
$p_h^{STM_{max}}$	Maximum power of battery storage h [MW]
$p_{max_b}^D$	Maximum active power of existing dispatchable generation units, bus b [MW]
$q_{max_b}^D$	Maximum reactive power of existing dispatchable generation units, bus b [MVar]
$p_{max_{vp}}^{VPD}$	Maximum active power of VPP dispatchable generation, vpp vp [MW]
$q_{max_{vp}}^{VPD}$	Maximum reactive power of VPP dispatchable generation, vpp vp [MVar]

p_{vp}^{VPR}	Maximum active power of VPP dispatchable generation, vpp vp [MW]
q_{vp}^{VPR}	Maximum reactive power of VPP dispatchable generation, vpp vp [MVar]
p_b^{CD}	Maximum active power of candidate dispatchable generation, bus b [MW]
q_b^{CD}	Maximum reactive power of candidate dispatchable generation, bus b [MVar]
p_b^{FxU}	Maximum active power of upward flexibility, bus b [MW]
p_b^{FxD}	Maximum active power of downward flexibility, bus b [MW]
$vref$	Reference bar for voltage angle
M	Large power value [p.u.]
Decision Variables	
$f_{l,t,s,w}$	Active power flow of line l , time period t , demand stage s , scenario w [MW]
$f_{l,t,s,w}^{Sign1}$	Signed active power flow of origin side of line l , time period t , demand stage s , scenario w [MW]
$f_{l,t,s,w}^{Sign2}$	Signed active power flow of destination side of line l , time period t , demand stage s , scenario w [MW]
$fQ_{l,t,s,w}$	Reactive power flow of line l , time period t , demand stage s , scenario w [MVar]
$p_{vp,b,t,s,w}^{VPR}$	Active power of VPP demanded from reserve market, vpp vp , bus b , time period t , demand stage s , scenario w [MW]
$q_{vp,b,t,s,w}^{VPR}$	Reactive power of VPP demanded from reserve market, vpp vp , bus b , time period t , demand stage s , scenario w [MVar]
$p_{vp,b,t,s,w}^{VPD}$	Active power of VPP dispatchable generation, vpp vp , bus b , time period t , demand stage s , scenario w [MW]
$q_{vp,b,t,s,w}^{VPD}$	Reactive power of VPP dispatchable generation, vpp vp , bus b , time period t , demand stage s , scenario w [MVar]
$p_{b,t,s,w}^D$	Active power of existing dispatchable generation units, bus b , time period t , demand stage s , scenario w [MW]
$q_{b,t,s,w}^D$	Reactive power of existing dispatchable generation units, bus b , time period t , demand stage s , scenario w [MVar]
$p_{b,t,s,w}^{CD}$	Active power of candidate dispatchable generation units, bus b , time period t , demand stage s , scenario w [MW]
$q_{b,t,s,w}^{CD}$	Reactive power of candidate dispatchable generation units, bus b , time period t , demand stage s , scenario w [MVar]
$p_{b,t,s,w}^{ND}$	Active power of existing non-dispatchable generation units, bus b , time period t , demand stage s , scenario w [MW]
$q_{b,t,s,w}^{ND}$	Reactive power of existing non-dispatchable generation units, bus b , time period t , demand stage s , scenario w [MVar]
$p_{vp,b,t,s}^{VPND}$	Active power of VPP non-dispatchable generation, vpp vp , bus b , time period t , demand stage s [MW]
$q_{vp,b,t,s}^{VPND}$	Reactive power of VPP non-dispatchable generation, vpp vp , bus b , time period t , demand stage s [MVar]
$p_{b,t,s,w}^{NDC}$	Active power of curtailed non-dispatchable generation units, bus b , time period t , demand stage s , scenario w [MW]
$p_{b,t,s,w}^{FxU}$	Active power of upward flexibility, bus b , time period t , demand stage s , scenario w [MW]
$p_{b,t,s,w}^{FxD}$	Active power of downward flexibility, bus b , time period t , demand stage s , scenario w [MW]
$dP_{FxU_{b,t,s,w}}^{RSP}$	Active power of procured demand response upward flexibility, bus b , time period t , demand stage s , scenario w [MW]

$dP_Fx D_{b,t,s,w}^{RSP}$	Active power of procured demand response downward flexibility, bus b , time period t , demand stage s , scenario w [MVar]
$p_{b,t,s,w}^{LC}$	Active power of curtailed demand, bus b , time period t , demand stage s , scenario w [MW]
$p_{h,t,s,w}^{STD}$	Storage active power discharge of storage h , time period t , demand stage s , scenario w [MW]
$p_{h,t,s,w}^{STC}$	Storage active power charge of storage h , time period t , demand stage s , scenario w [MW]
$pDur_s$	Time duration of demand stage s [pu]
$V_{b,t,s,w}$	Voltage (p.u.) at bus b at time t , demand stage s , scenario w
$\delta V_{b,t,s,w}$	Voltage (p.u.) at bus b at time t , demand stage s , scenario w
$\theta_{ij,t,s,w}$	Voltage phase angle between nodes i and j at time t , demand stage s , scenario w
$SoC_{h,t,s,w}$	State-of-charge (MW h), storage h , at time t , demand stage s , scenario w
$ESS_{h,t,w}$	Energy available at storage device h in time period t , scenario w [MW h]
$admitG_l$	Admittance of line l - real part
$admitB_l$	Admittance of line l - imaginary part
$iFx D_{b,s}$	Binary variable indicating if downward flexibility is considered at bus b , during demand stage s [0,1]
$iFx U_{b,s}$	Binary variable indicating if upward flexibility is considered at bus b , during demand stage s [0,1]
$Circ_{l,c,t}$	Binary variable indicating the presence of a circuit c , in corridor l , time period t [0,1]
$InST_{h,t}$	Binary variable indicating the presence of storage h , time period t [0,1]
$InVL_{l,t}$	Binary variable indicating the presence of VPL l , time period t [0,1]
$ip_{cd,b,t}^{CD}$	Binary variable indicating the presence of candidate dispatchable generation cd , bus b , time period t [0,1]
$ip_{cnd,b,t}^{CND}$	Binary variable indicating the presence of candidate non-dispatchable generation cnd , bus b , time period t [0,1]
$soc_{h,t,s}$	Binary variable indicating the charge/discharge status of battery storage unit h , time period t , stage s [0,1]
$VPL_Fij1_St_{l,t,s,w}$	Binary variable indicating the flow status of VPL line l from side, time period t , stage s [0,1]
$VPL_Fij2_St_{l,t,s,w}$	Binary variable indicating the flow status of VPL line l to side, time period t , stage s [0,1]
$VPL_Carg1_St_{l,t,s,w}$	Binary variable indicating the charge/discharge status of VPL Battery storage unit 1 of line l , time period t , stage s [0,1]
$VPL_Carg2_St_{l,t,s,w}$	Binary variable indicating the charge/discharge status of VPL Battery storage unit 2 of line l , time period t , stage s [0,1]
Vector Notation	
y	Set of network variables: $[Circ_{l,c,t,w}, InST_{h,t,w}, ip_{cd,b,t,w}^{CD}, ip_{cnd,b,t,w}^{CND}]$ $\forall b \in \Omega^B, h \in \Omega^H, t \in \mathcal{T}, s \in \Omega^S, v \in \Omega^V, w \in \Omega^W$.
x	Set of network variables: $[p_{v,b,t,s,w}^{VPR}, q_{v,b,t,s,w}^{VPR}, p_{v,b,t,s,w}^{VPD}, q_{v,b,t,s,w}^{VPD}, p_{v,b,t,s,w}^{VPND}, q_{v,b,t,s,w}^{VPND}, p_{b,t,s,w}^D, q_{b,t,s,w}^D, p_{b,t,s,w}^{CD}, q_{b,t,s,w}^{CD}, p_{b,t,s,w}^{ND}, q_{b,t,s,w}^{ND}, p_{b,t,s,w}^{CND}, q_{b,t,s,w}^{CND}, p_{b,t,s,w}^{NDC}, q_{b,t,s,w}^{NDC}, p_{b,t,s,w}^{FXU}, q_{b,t,s,w}^{FXU}, p_{b,t,s,w}^{FXD}, q_{b,t,s,w}^{FXD}, iFx D_{b,s,w}, iFx U_{b,s,w}, dP_{b,t,s,w}, dQ_{b,t,s,w}, dP_{b,t,s,w}^{RSP}, dQ_{b,t,s,w}^{RSP}, p_{b,t,s,w}^{LC}, q_{b,t,s,w}^{LC}, p_{h,t,s,w}^{STD}, q_{h,t,s,w}^{STD}, p_{h,t,s,w}^{STC}, q_{h,t,s,w}^{STC}, v_{b,t,s,w}, \theta_{ij,t,s,w}]$ $\forall b \in \Omega^B, h \in \Omega^H, t \in \mathcal{T}, s \in \Omega^S, v \in \Omega^V, w \in \Omega^W$.

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